Data and Process Modeling for Restaurant Ordering System

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System Overview

The restaurant ordering system is designed to create an efficient dining experience for both customers and staff. The system provides a digital, hybrid platform, accessible through tablets at tables where customers can browse the menu, place orders, and request services without waiting for staff. The system manages everyday restaurant operations, including menu presentation, table management, order processing, and payment handling.

Management can also use this system to track inventory, monitor sales, and generate reports for business insights. Overall, this system improves efficiency, reduces wait times, minimizes human error in stressful rushes, and enhances customer experience while giving restaurant management ease of operation as well as better control.

Functional Decomposition Diagram

The restaurant ordering system will be a multiplatform application, which includes a mobile application (IOS, ANDROID), website, and desktop app.

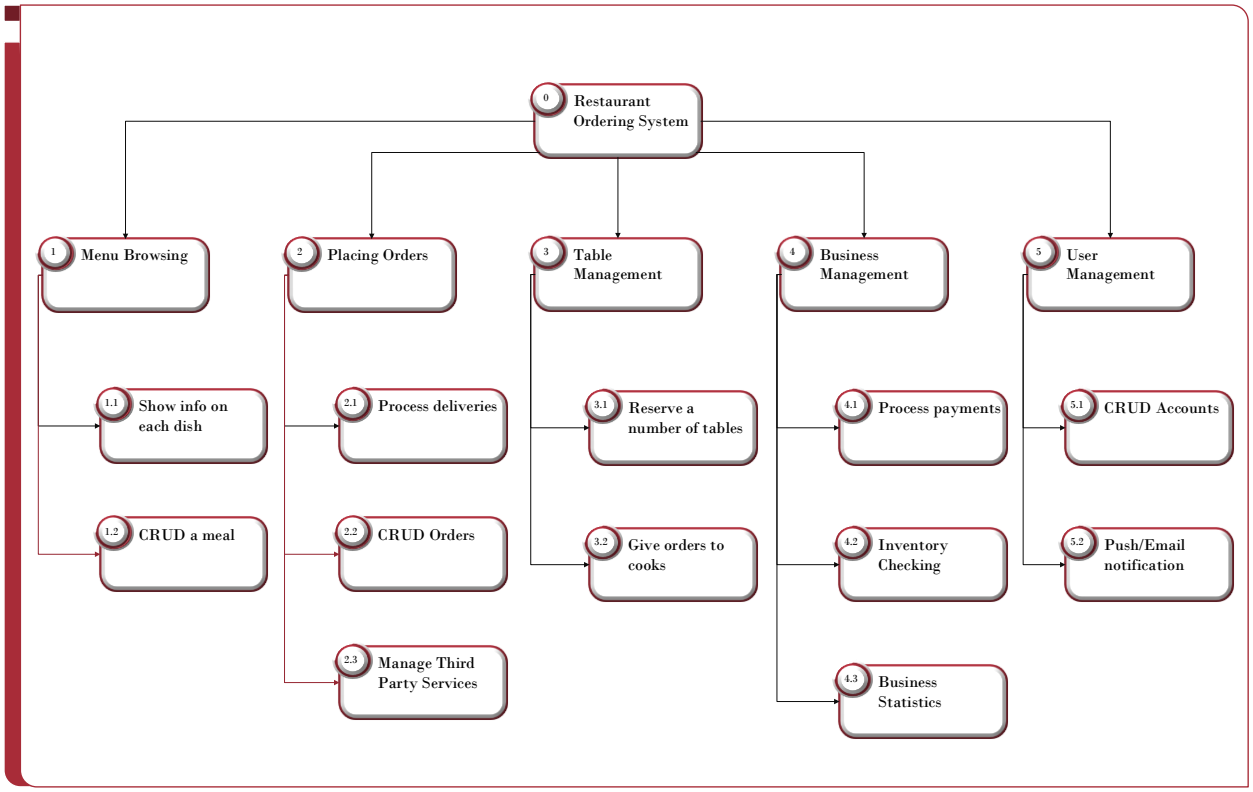
The ordering system will perform many functions/automations for both the customer, workers, and owner.

The system needs to handle many CRUD operations (Create, Read, Update, Delete), such as the menu, accounts, and orders.

The system needs to handle business logic for the owner, such as inventory management, payments, and business statistics.

The system will need extra logic for giving orders for chefs and maintaining a reservations schedule.

The app will list information in the menu, such as estimated time to make, cost, nutrition facts, description, and availability.



Context Diagram

The restaurant ordering system will interact with five entities, the customer, employees, the owner payment processor, and delivery app.

First, starting with the customer, they will need to interact with a user interface, place orders, reserve tables, create and modify an account.

The employees will receive an order to cook the food and alert the system when it is ready. The owner will receive some business analytics including statistics of how the business is going, and they will be able to manage/modify the menu and communicate with the employees.

The payment would include sending and confirming whether a person has paid; it will have to include many different payment types.

The last entity is the delivery app; the system will interact with it, to manage the delivery of the food. Notable third-party delivery apps include DoorDash, Uber Eats, and Grubhub.

Important terminology

Dish: A food/drink item available in the restaurant
Menu: a list of dishes
Table: Reservable amount seats, within a restaurant
Delivery: Sending food to a location like a home/apartment
Take-out: Customers receive their food, and eat somewhere else
Order: A chosen set of dishes from the menu, from the customer; it is complemented with either a table, delivery, or takeout.

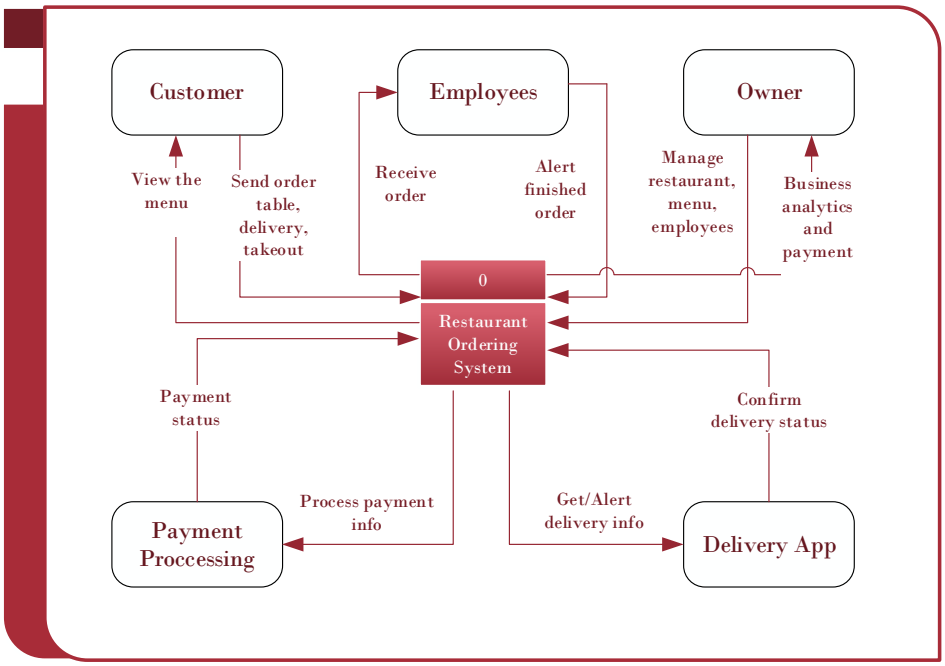


Diagram 0

Process 1 Menu Browsing: Query information about what the restaurant has. Includes info about availability, prices, nutrition, and estimated time to get ready.

Process 2 Placing Orders: From the pick-out menu, an order will be queued for the chefs to make and then queued for delivery.

Process 3 Table Management: Allows users to reserve a pre-paid table soon, accounts for the number of people and places to order.

Process 4 Business Management: gathers statistics on the

Process 5 User Management: Allows CRUD accounts, set up email/push notifications, set up payment and have access to the restaurant's services

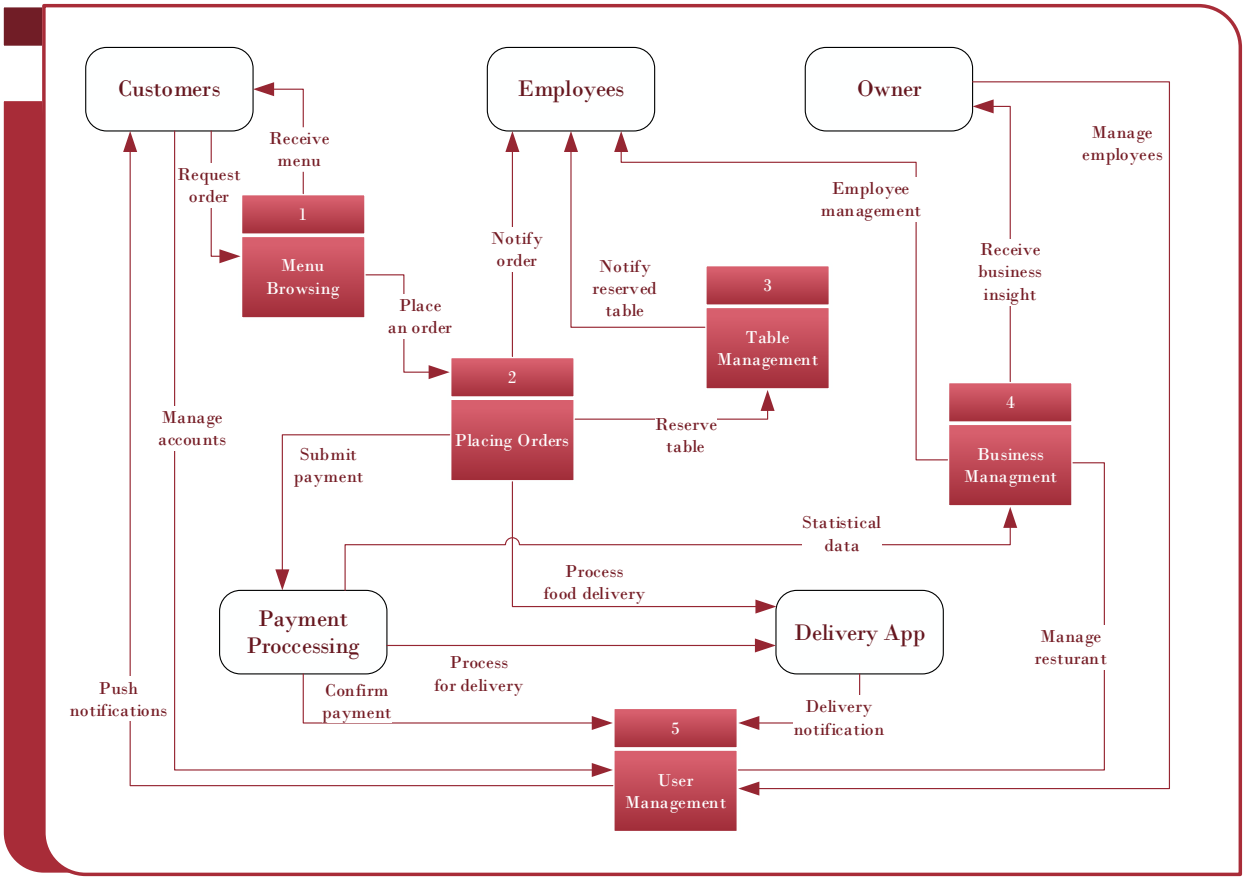


Diagram 1

Process 1.1

Input/Output Customer: Takes in a query, either a list of everything on the menu or specific info on an item. General info would be the name, short description, price, main ingredients, and time to make. The specific info would be all nutritional information, and full description.

Process 1.2

Input from Customer: Takes in a list of meals the user has chosen, along with their option of a table, takeout or delivery. It also would make a request to edit or cancel an order.

Output to Customer: Gives confirmation to the customer about their order, whether it has been set, finished, modified or cancelled.

Output to Placing orders (2): Sends the processed data from the order, which includes information about the meals, customer name, payment, type of order (table, take-out, or delivery).

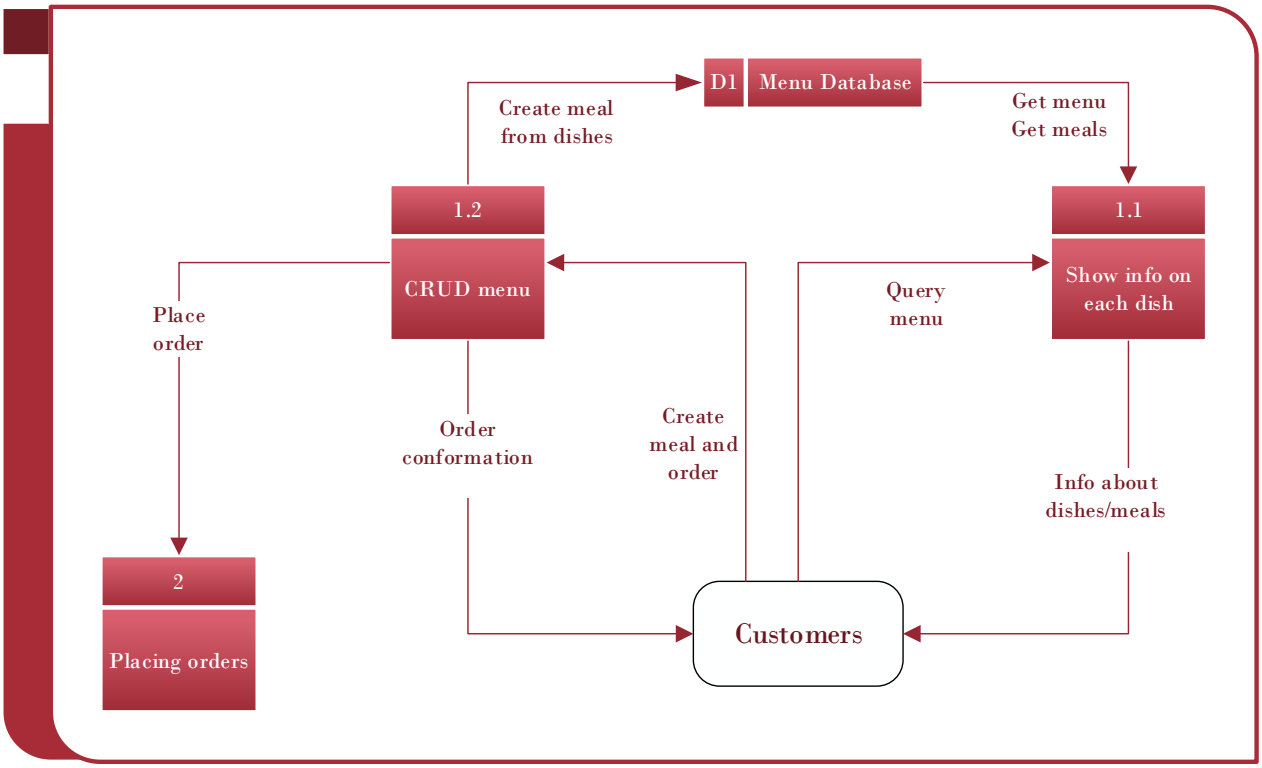


Diagram 2

Process 2.1

Input: Delivery Info from CRUD Orders (2.2): If an Order contains Delivery Info, it will be processed here into a manageable format for the Delivery App.

Output: Delivery Info to Manage Third Party Services (2.3): That process will handle sending the Deliver Info to the Delivery App.

Process 2.2

Inputs:

- Order Info from Received Orders (D1): Orders Placed by Customers from Menu Browsing (1) are sent to the Received Orders data store, before being processed here to be sent to other processes.
- Completed Order from Orders to Make (D2): After Employees complete or cancel an order, this process will either send the completed order to Table Reservations (3) or delete them.
- Payment Confirmation from Manage Third Party Services (2.3): The confirmation of payment will notify the process of sending the unprepared Order to the Employees to make.

Outputs:

- Table Reservation to Table Management (3): Includes information about how many Customers will need to be seated.
- Orders to Make to Orders to Make (D2): Sends the orders off to the Employees to prepare them.
- Delivery Info to Process Deliveries (2.1).

Process 2.3

Inputs:

- Payment Info from Received Orders (D1). Contains payment information from the Customer's orders, encrypted of course.
- Delivery Info from Process Deliveries (2.1). Contains information about the delivery of the order, such as the contents of the order, the customer's address, and priority.
- Payment Confirmed from Payment Processing (E). Informs the process if the payment from the Customer has succeeded or failed. It is not just a confirmation of success.

Outputs:

- Payment Submitted to Payment Processing (E). Send Customer payment information over to the third-party payment processor.
- Payment Confirmation to CURD Orders (2.2). Informs the order management process whether the payment had succeeded or not.
- Delivery Info to Delivery App (E). Send information about the delivery to the app. This includes information for the driver (customer address, proper handling of order, etc.) and for the customer (order confirmation, estimated time of delivery)

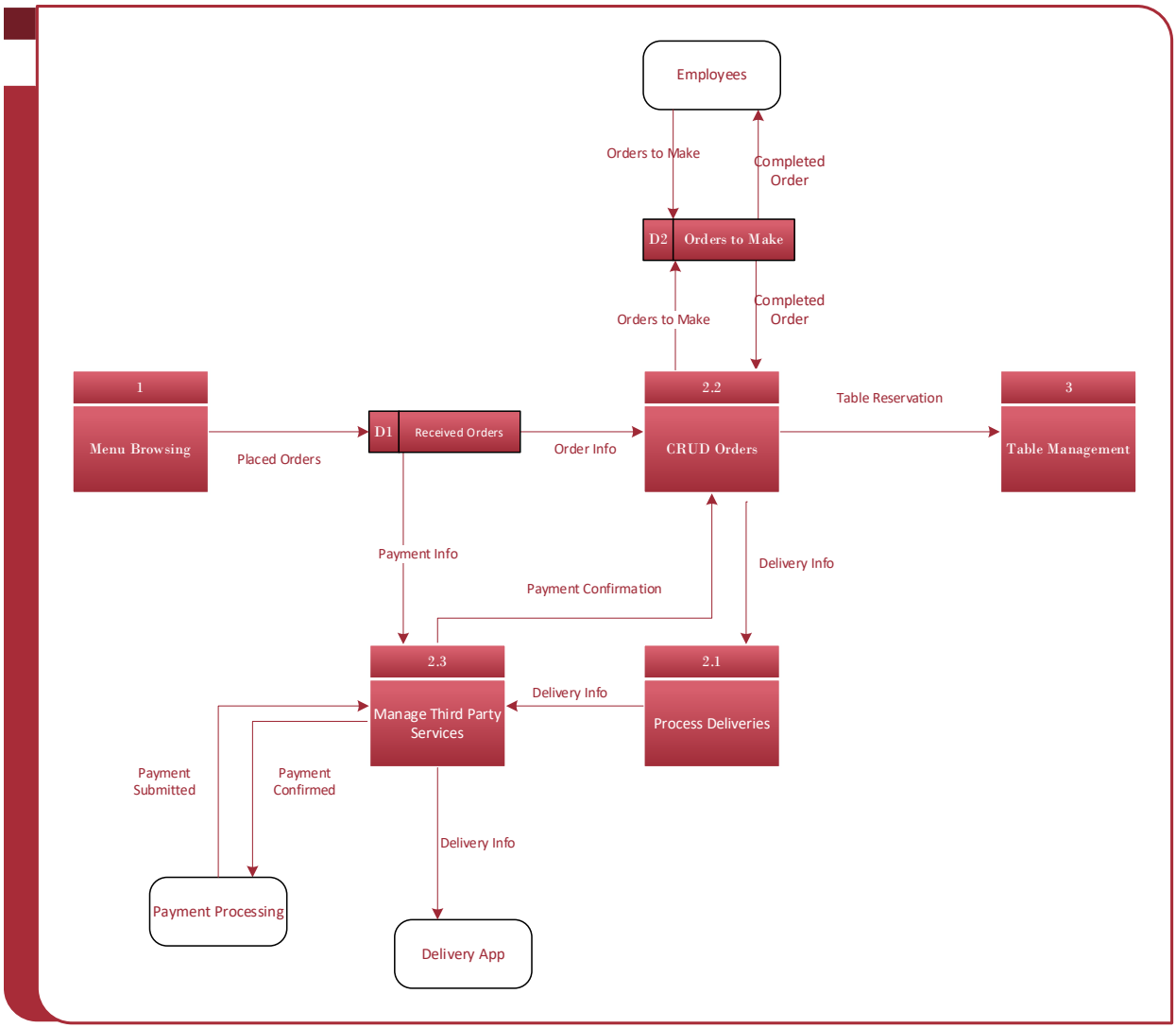


Diagram 3

Process 3.1

Input: “Successful Reservation” from Status of Restaurant Tables (D1): This prompt tells the process to book a requested table for the number of Customers in the party.

Outputs:

- “Failed Reservation/Table Empty” to D1: If a “reservation could not be made or a” table is empty, it updates that table’s information in the data store.
- Table w/ Customers to Orders to Cooks & Waiter (3.2). Names of customers.

Process 3.2

Input: Table w/ Customers from Reserve a Number of Tables (3.1)

Output: Notification about where customers are seated to Employees (E): Links orders and customers together so Employees will know where the Customers are seated so their orders can be delivered.

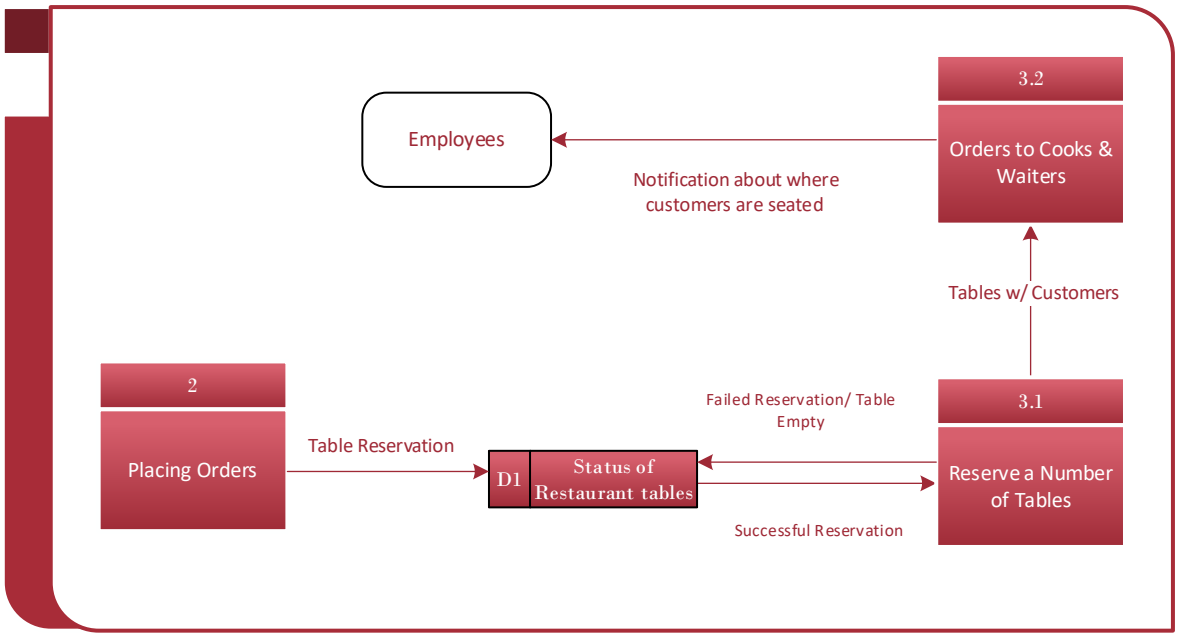


Diagram 4

Process 4.1

- Receives information on current inventory and what we need to buy as well as the current budget we have to work with.

Process 4.2

- Owner entity starts the process by checking the inventory which pulls from the Inventory database, updates the current inventory, and returns the new inventory. It then buys the missing inventory outlined in the new inventory within budget.

Process 4.3

- Owner entity starts the process by opening business statistics which checks financials through the financials database and then sends that information to

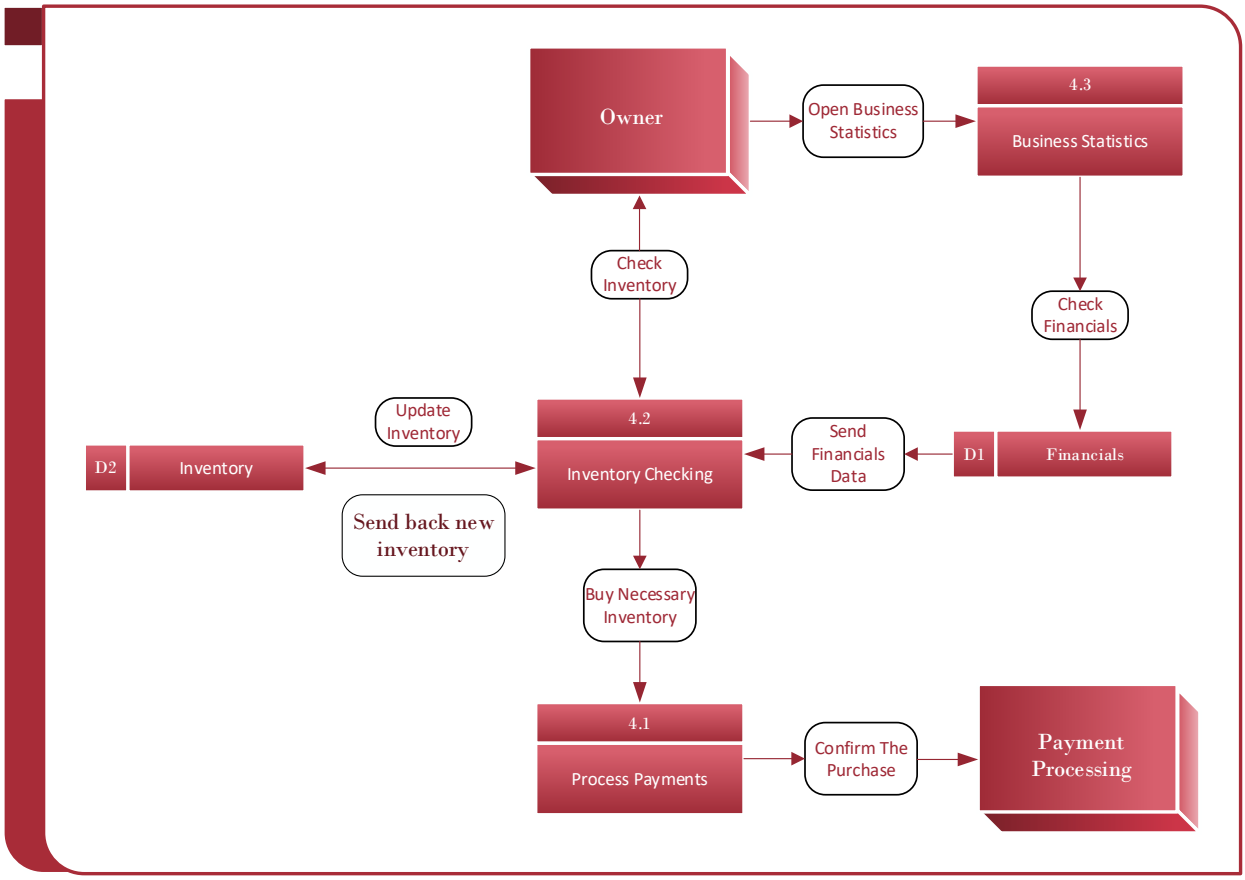


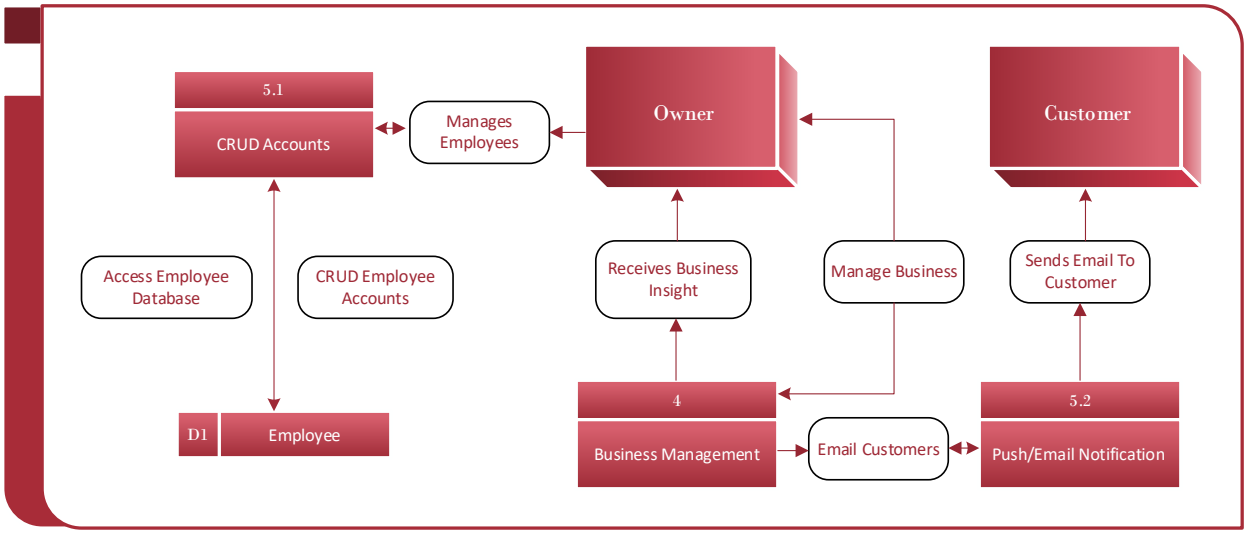
Diagram 5

Process 5.1

- This process will be managed by the owner who manages the employees by CRUD. When an account is CRUD, it will access the employee database to update the relevant information and then send back the changes.

Process 5.2

- The business will email customers about appointments, confirmations, deliveries, etc, and the process will make sure the email reaches the customer.



Functional Primitives

1.1 Show Info On Each Dish

RECEIVE Menu Query from Customer
IF Query = “Full Menu” THEN
SEND List of Meals (name, description, price, ingredients, prep time) to Customer
ELSE IF Query = “Specific Item” THEN
SEND Detailed Meal Info (nutritional info, full description) to Customer

1.2 CRUD a Meal/Customer Order Input

RECEIVED Order Details from Customer
GET Selected Meals from Order Details
GET Order Type from Order Details (table, takeout, delivery)
IF Customer Creates Order THEN
SEND Order Confirmation to Customer
SEND Order Info (meals, name, payment, type) to Placing Orders (2)
ELSE IF Customer Edits Order THEN
SEND Modified Confirmation to Customer
SEND Updated Order Data to Placing Orders (2)
ELSE IF Customer Cancels Order THEN
SEND Cancellation Confirmation to Customer
SEND Cancellation Request to Placing Orders (2)

2.1 Process Deliveries

RECEIVE Delivery Info from CRUD Orders (2.2)
IF Order Contains Delivery Details THEN
FORMAT Delivery Info for Delivery App
SEND Delivery Info to Manage Third Party Services (2.3)

2.2 CRUD Orders

RECEIVE Order Info from Received Orders (D1)
RECEIVE Completed Orders from Orders to Make (D2)
RECEIVE Payment Confirmation from Manage Third Party Services (2.3)
IF Payment Confirmed THEN

SEND Unprepared Order to Orders to Make (D2)
ELSE
DELETE Cancelled Orders
IF Table Order THEN
SEND Reservation Info to Table Management (3)
IF Delivery Order THEN
SEND Delivery Info to Process Deliveries (2.1)
IF Order Completed THEN
STORE Completed Order in Completed Orders (D3)

2.3 Manage Third Party Services

RECEIVE Payment Info from Received Orders (D1)
RECEIVE Delivery Info from Process Deliveries (2.1)
RECEIVE Payment Status from Payment Processing
IF Payment Needed THEN
SEND Payment Data to Payment Processing
IF Payment Confirmed THEN
SEND Payment confirmation to CRUD Orders
IF Delivery Required THEN
SEND Delivery Details to Delivery App

3.1 Reserve a Number of Tables

RECEIVE Reservation Request from Table Management (30
RECEIVE Table Status from Status of Restaurant Tables (D1)
IF Table Available THEN
STORE Reservation in Table Database
SEND Table with Customer Info to Give Orders to Cooks (3.2)
ELSE
UPDATE Table Status to “Unavailable” in D1

3.2 Give Order to Cooks

RECEIVE Table with Customers from Reserve a Number of Tables (3.1)
LINK Orders with Table and Customers
SEND Seating Notification (table number, order link) to Employees

4.1 Process Payments

RECEIVE Current Inventory and Budget from Inventory Database
IF Inventory or Budget Updated THEN
SEND Purchase Request to Inventory Checking (4.2)

4.2 Inventory Checking

RECEIVE Inventory Data from Inventory Database
GET Current Stock from Inventory Data
IF Items missing AND Within Budget THEN
SEND Purchase Orders for missing items
UPDATE Inventory Database with New Stock

4.3 Business Statistics

RECEIVE Request from Owner to View Statistics
GET Financial Data from Financials Database
SEND FINANCIAL Data from Financials Database
SEND Business Summary Report to Owner

5.1 CRUD Accounts

RECEIVE Account Update Request from Owner
GET Employee Info from Employee Database
IF Account Exists THEN
UPDATE Employee Record
ELSE
STORE New Employee Record
SEND Account Update Confirmation to Owner

5.2 Push/Email Notification

RECEIVE Notification Data (appointments, confirmations, deliveries)
IF Notification Valid THEN
SEND Email Notification to Customer
ELSE
    REJECT Notification Request

SEND Failure Notice to Business

Conclusion

This restaurant ordering system represents a comprehensive approach to improving efficiency and customer satisfaction with a dining environment. By developing key processes like menu browsing, order placement, table management and inventory reviews, all onto a digital platform: this system will minimize errors, wait times, and improve the restaurant experience. Our usage of data flow diagrams, functional decomposition, and process descriptions amount to a platform that improves front and back-end restaurant operations. These designs allow for scalable versions that balance convenience and leadership oversight. This ensures restaurants can take advantage of these models for any in-person or delivery needs.